

Submission Summary

Conference Name

IEEE International Conference on Computer, Electrical, and Electronics Engineering

Paper ID

1127

Paper Title

Digital Forensics in Cybersecurity: Techniques for Effective Evidence Investigation

Abstract

The digital forensics process is vital in today's cybersecurity landscape since it facilitates the identification, collection, preservation, and analysis of digital evidence in cybercrime cases. In the face of rising digital technologies and threats like malware attacks, data breaches, and access violations, it is imperative to have efficient forensic investigation techniques. This research analyzes various methodologies and tools used for advanced digital forensics and their impact on improving digital investigation. An experiment involving simulated cyberattacks was conducted within the framework of virtual machines. The tools like Autopsy and Wireshark were used in collecting and analyzing evidence in the experimental setting. Furthermore, techniques of automation using scripting languages like Python and Bash scripts were employed to speed up the investigation process. The findings show that a well-structured process of digital forensics coupled with automation improves the efficiency of the evidence collection and analysis.

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Primary Subject Area

Computer Engineering

Secondary Subject Areas

Cyber Security

Submission Files

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